ZSWatch/ZSWatch

ZSWatch - the Open Source Zephyr™ based Smartwatch, including both HW and FW.

April 2026

Executive Summary

ZSWatch is an innovative open source smartwatch project, encompassing both hardware and firmware, built upon the robust Zephyr RTOS and Nordic Semiconductor nRF5340 SoC. The project champions transparency and hackability in the wearable technology space, offering a comprehensive development kit (WatchDK) and a rich set of features from advanced sensors to modular applications. Partially funded by the NGI0 Commons Fund, ZSWatch is positioned to serve a growing market of developers, privacy conscious users, and open source hardware enthusiasts seeking full control and customization over their wearable devices.

8/100
OVERALL SCORE

7
MARKET FIT

9.5
TECHNICAL

ZSWatch: The Open Source Smartwatch

ZSWatch: A fully open source smartwatch platform, combining innovative hardware with the robust Zephyr RTOS. Empowering developers and users with transparency and customization.

Executive Speaker Notes:
Good morning. We are here to introduce ZSWatch, a groundbreaking open source smartwatch project. Our mission is to redefine wearable technology by providing a platform that is completely open, from its circuit boards to its firmware, built on the Zephyr Real Time Operating System. This is not just another smartwatch; it's a foundation for innovation and user empowerment.

The Closed Smartwatch Ecosystem Problem

1. Vendor lock-in limits user and developer choice. 2. Lack of transparency raises privacy and security concerns. 3. High barriers to entry for hardware and firmware customization stifle innovation. 4. Existing solutions fail to meet the needs of the open source and privacy conscious communities.

Executive Speaker Notes:
Today's smartwatch market, dominated by a few large corporations, presents significant challenges. Users are locked into proprietary ecosystems, with no real control over their devices. Developers face immense barriers trying to innovate on these closed platforms. This lack of transparency leads to privacy concerns, as users have little insight into how their data is handled. We believe there's a better way, a more open and trustworthy way.

ZSWatch: Your Open and Customizable Wearable

1. Fully open source hardware designs, enabling complete transparency and DIY builds. 2. Zephyr RTOS based firmware, offering a secure, efficient, and modifiable software stack. 3. Modular application framework with LVGL UI for endless customization. 4. Development Kit (WatchDK) provides an easy entry point for developers and hobbyists. 5. Cross platform mobile companion apps for seamless connectivity and notification management.

> **Executive Speaker Notes:**
> Our solution is ZSWatch, the open source smartwatch. We provide all hardware designs in KiCad, allowing anyone to build or modify the device. Our firmware, built on Zephyr RTOS, is highly modular, secure, and energy efficient. The user interface, powered by LVGL, is designed for easy customization and new application development. For immediate engagement, our WatchDK development kit is available, simplifying development. We also offer companion apps for iOS and Android, ensuring broad compatibility.

Untapped Market for Open Source Wearables

1. Growing demand from open source hardware and software enthusiasts. 2. Niche market of privacy conscious consumers seeking transparent technology. 3. Educational institutions and IoT developers for prototyping and learning. 4. Industrial applications requiring highly customizable and robust wearable solutions. 5. European Commission's NGI0 program validation indicates strategic market relevance.

Executive Speaker Notes:
ZSWatch targets a significant yet underserved market. This includes the vibrant open source community, individuals deeply concerned about digital privacy, and embedded systems developers who need flexible, hackable hardware for prototyping or specialized industrial use cases. The backing from the NGI0 Commons Fund underscores the strategic importance of open, trustworthy hardware in Europe and beyond.

Robust Technology Stack

1. Hardware: nRF5340 SoC, 240×240 round touch display, comprehensive sensor suite (IMU, pressure, magnetometer, light, mic, RTC), 64 MB external flash, and modular HR extension. All KiCad designed. 2. Firmware: Zephyr RTOS (NCS v3.1.0) written in C, featuring a modular app framework, Zbus event system, and LVGL v9 UI with XML based design. 3. Connectivity: Bluetooth LE with support for GadgetBridge, Apple ANCS/AMS/CTS, HID, and HTTP proxy. 4. Development Tools: CMake, West, VS Code integration, Pytest for HIL and native simulation testing, CI/CD pipelines, and web based firmware update. 5. Power Management: Optimized for low power with Nordic nPM1300 PMIC and execute in place (XIP) from external flash.

> **Executive Speaker Notes:**
> Technically, ZSWatch is advanced. Our hardware features the powerful nRF5340 SoC, a crisp round touch display, and a rich array of sensors for activity tracking, environment monitoring, and timekeeping. The modular heartrate extension is also under development. The firmware is built on Zephyr RTOS, known for its small footprint, security, and real time capabilities. We utilize LVGL for a responsive and extensible user interface. Our robust BLE stack enables connectivity with various mobile devices and services. Our development tools, including Pytest for comprehensive testing and a web based firmware update utility, streamline the development process. Power consumption is meticulously managed for extended battery life.

Sustainable Open Source Business Model

1. Sales of WatchDK development kits via Elecrow. 2. Professional services: Custom development, integration, and consulting for enterprises. 3. Future direct sales of assembled ZSWatch devices. 4. Community funding: Grants and donations supporting continuous R&D. 5. Potential for a marketplace for certified open source applications and value added services. The GPL-3.0 license fosters community contributions and collaborative improvement.

> **Executive Speaker Notes:**
> Our business model is designed for sustainability within the open source paradigm. We currently generate revenue from selling the WatchDK development kits. In the future, we foresee expanding into professional services for companies needing custom wearable solutions. We will also sell fully assembled ZSWatch devices. Our foundation is strengthened by grants, such as the NGI0 Commons Fund, and community donations. We can also explore a marketplace for open source apps and premium support tiers. The GPL-3.0 license is central to our philosophy, ensuring that all innovations benefit the entire community.

Unique Strengths in a Crowded Market

1. Unparalleled Transparency: First truly open source smartwatch hardware and software, fostering trust and hackability. 2. Developer Ecosystem: Built on a modern, well supported embedded stack (Zephyr RTOS, NCS) with dedicated developer tools. 3. Privacy Focused: Openness naturally addresses privacy concerns inherent in proprietary wearables. 4. Modular Design: Highly extensible for custom applications, watchfaces, and future hardware modules. 5. Community Backed: Active community and European Union funding validate the project's vision and provide a strong foundation.

Executive Speaker Notes:
Our competitive edge lies in our commitment to openness. Unlike any other smartwatch, ZSWatch offers complete transparency and hackability. This appeals to a growing segment of developers and privacy conscious users. Our foundation on Zephyr RTOS and Nordic's SDK provides a state of the art development environment. The modular architecture ensures that ZSWatch can evolve and adapt, not just through our efforts but through community contributions. Our active community and critical funding validate our potential for long term impact.

Addressing Potential Challenges

1. Hardware Scaling: Transitioning to mass production from development kits is complex and capital intensive. 2. Market Adoption: Niche appeal may limit widespread consumer market penetration against established giants. 3. Feature Parity: Keeping pace with rapid advancements in commercial smartwatches requires consistent R&D. 4. Funding Sustainability: Diversifying funding beyond grants for long term project viability. 5. Ecosystem Growth: Building a vibrant third party app ecosystem can be challenging. We are actively mitigating these risks through careful planning, community engagement, and strategic partnerships, focusing on our core strengths of openness and developer empowerment.

Executive Speaker Notes:
As with any ambitious project, we face challenges. Scaling hardware manufacturing and securing reliable supply chains are complex. Mainstream market adoption might be slow due to the inherent niche nature of open source products. We must continuously innovate to maintain feature parity with rapidly evolving commercial offerings. Sustaining long term funding beyond initial grants is critical. Finally, building a thriving developer ecosystem requires ongoing effort. We are actively addressing these risks by focusing on our core community, strategic partnerships, and a clear roadmap for both hardware and software development.

Join the ZSWatch Revolution

We invite you to join the ZSWatch community. 1. Developers: Contribute code, build applications, or create new hardware modules. 2. Enthusiasts: Engage with our community, provide feedback, and help us grow. 3. Partners: Collaborate on hardware manufacturing, distribution, or enterprise solutions. Visit zswatch.dev to learn more, access documentation, and get involved. Together, we can build the future of open wearable technology.

> **Executive Speaker Notes:**
> We believe ZSWatch represents a significant step towards truly open and customizable wearable technology. We invite you to be a part of this revolution. Whether you are a developer looking for a platform to build on, an enthusiast who values transparency, or a potential partner interested in shaping the future of wearables, ZSWatch offers a unique opportunity. Join us at zswatch.dev and help us redefine what a smartwatch can be.